

**Direct edge-to-map: actions keeping map age within budget
for at least 50% of route time (counterfactual)**

Network profile

Favorable Stable

0/72
best 10.2%

0/72
best 39.5%

22/72
best 70.1%

43/72
best 88.0%

Mid Variable

0/72
best 8.0%

0/72
best 35.7%

15/72
best 68.1%

36/72
best 86.7%

Adverse Stable

0/72
best 8.6%

0/72
best 36.3%

13/72
best 69.2%

24/72
best 87.6%

Fade Recovery

0/72
best 8.0%

0/72
best 36.5%

13/72
best 68.5%

38/72
best 86.3%

150 ms

200 ms

250 ms

300 ms

Physical map-freshness budget

Qualifying actions (of 72)

70

60

50

40

30

20

10

0